NEPAL COLLEGE OF INFORMATION AND TECHNOLOGY

Balkumari, Lalitpur

Affiliated to Pokhara University

DATABASE MANAGEMENT SYSTEM

Laboratory Report

Lab Title	**Lab 3: Familiarization with SQL Constraints & Wildcard Characters**
Subject	Database Management System (DBMS)
Tool Used	XAMPP, phpMyAdmin, MySQL
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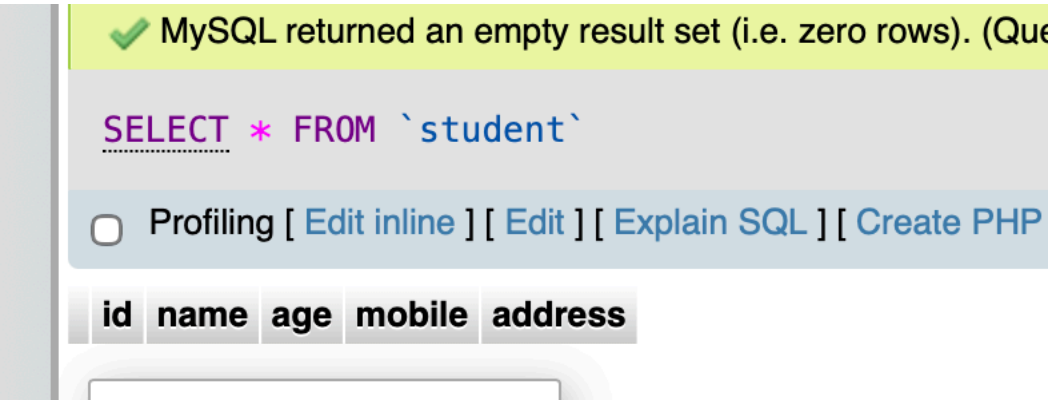
Roll No.	**241540**
Submitted To	**Er. Manil Vaidhya**
Date	**9rd Jun, 2026**

Department of IT Engineering

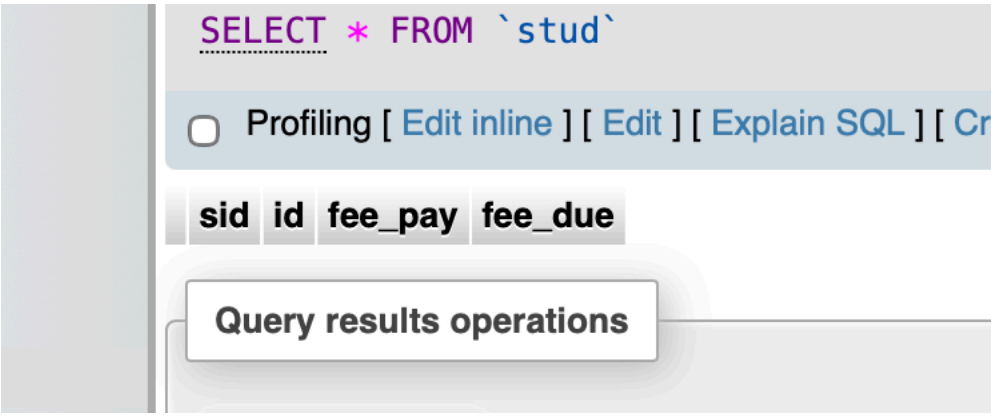
Q1. Create table student with constraints

```
CREATE TABLE student(
   id int primary KEY,
   name varchar(30) NOT NULL,
   age int NOT NULL check(age<20),
   mobile bigint UNIQUE NOT NULL,
   address varchar(30) DEFAULT 'KTM'
   )
```

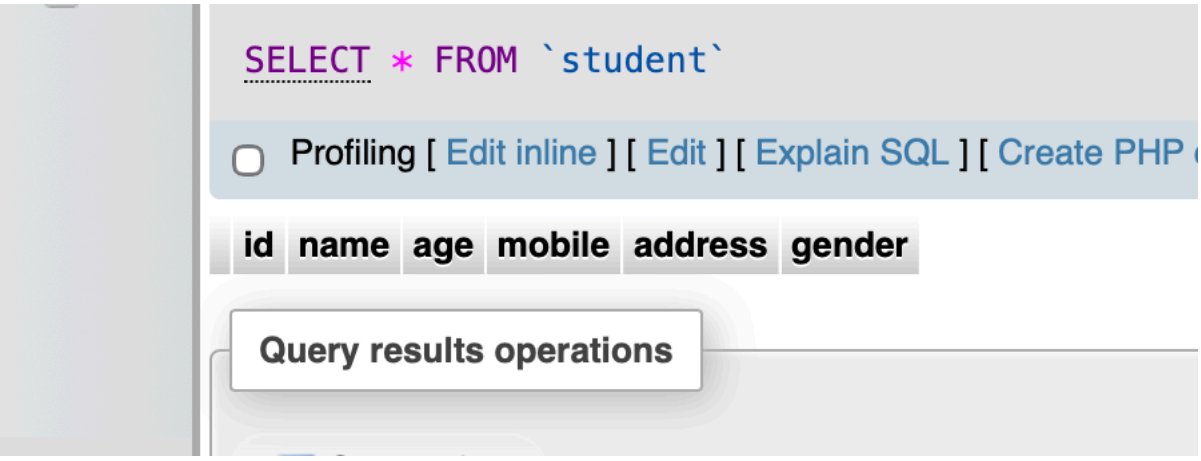


Q2. Create table stud with Foreign Key

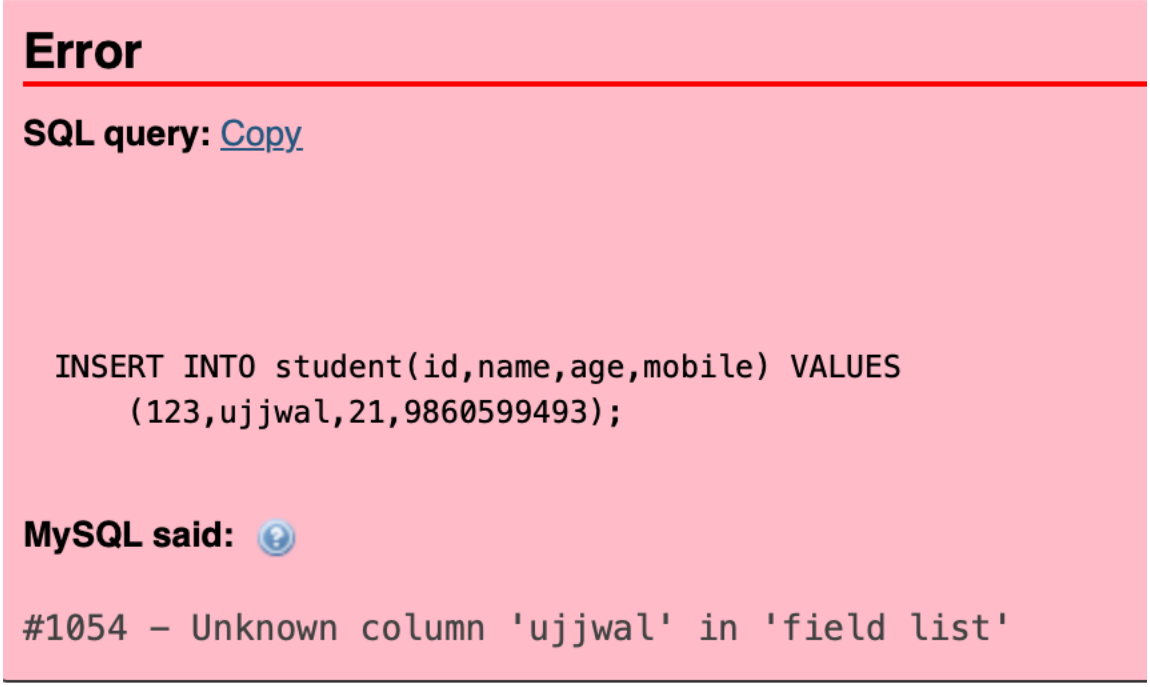
```
CREATE TABLE stud(
   sid int primary KEY,
   id int,
   fee_pay decimal,
   fee_due decimal,
   FOREIGN KEY (id) REFERENCES student(id)
   )
```



Q3. Add column gender with NOT NULL and DEFAULT
ALTER TABLE student ADD COLUMN gender varchar(30) NOT NULL DEFAULT 'Male';



Q4. Insert records (including invalid ones)
INSERT INTO student(id,name,age,mobile) VALUES
(123,ujjwal,21,9860599493);



```
INSERT INTO student(id,name,age,mobile,address) VALUES
   (123,'ujjwal',18,9860599493,'chitwan'),
   (124,'Ram',13,9860599494,'gaur'),
   (126,'ujjwal',19,9860599495,'bharatpur'),
   (125,'shyam',15,9860599496,'Rautahat'),
   (128,'ujjwal',12,9860599491,'pkr')
   ;
```

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: No

Extra options

	id	name	age	mobile	address	gender
Edit Copy Delete	123	ujjwal	18	9860599493	chitwan	Male
Edit Copy Delete	124	Ram	13	9860599494	gaur	Male
Edit Copy Delete	125	shyam	15	9860599496	Rautahat	Male
Edit Copy Delete	126	ujjwal	19	9860599495	bharatpur	Male
Edit Copy Delete	128	ujjwal	12	9860599491	pkr	Male

Check all *With selected:* Edit Copy Delete Export

```
INSERT INTO stud(sid,id,fee_pay,fee_due) VALUES
      (465,123,20000.674,10000.46),
   (466,126,20000.674,10000.46),
   (467,125,20000.674,10000.46);
```

Show all | Number of rows: 25 Filter rows: Search this tabl

Extra options

	sid	id	fee_pay	fee_due
Edit Copy Delete	465	123	20001	10000
Edit Copy Delete	466	126	20001	10000
Edit Copy Delete	467	125	20001	10000

Check all *With selected:* Edit Copy Delete

Q5. Display all table info

SELECT * FROM student,stud;

Extra options

id	name	age	mobile	address	gender	sid	id	fee_pay	fee_due
123	ujjwal	18	9860599493	chitwan	Male	465	123	20001	10000
123	ujjwal	18	9860599493	chitwan	Male	466	126	20001	10000
123	ujjwal	18	9860599493	chitwan	Male	467	125	20001	10000
124	Ram	13	9860599494	gaur	Male	465	123	20001	10000
124	Ram	13	9860599494	gaur	Male	466	126	20001	10000
124	Ram	13	9860599494	gaur	Male	467	125	20001	10000
125	shyam	15	9860599496	Rautahat	Male	465	123	20001	10000
125	shyam	15	9860599496	Rautahat	Male	466	126	20001	10000
125	shyam	15	9860599496	Rautahat	Male	467	125	20001	10000
126	ujjwal	19	9860599495	bharatpur	Male	465	123	20001	10000
126	ujjwal	19	9860599495	bharatpur	Male	466	126	20001	10000
126	ujjwal	19	9860599495	bharatpur	Male	467	125	20001	10000
128	ujjwal	12	9860599491	pkr	Male	465	123	20001	10000
128	ujjwal	12	9860599491	pkr	Male	466	126	20001	10000
128	ujjwal	12	9860599491	pkr	Male	467	125	20001	10000

Q6. Display distinct name

SELECT DISTINCT name FROM student,stud;

Extra options

name
ujjwal
Ram
shyam

Show all

Q7. Select name and id of 'Ram'

SELECT name ,id FROM student WHERE name='ram';

Extra options

	name	id
Edit Copy Delete	Ram	124

Extra options

	id	name	age	mobile	address	gender
Edit Copy Delete	123	ujjwal	18	9860599493	chitwan	Male
Edit Copy Delete	124	Ram	13	9860599494	gaur	Male
Edit Copy Delete	125	shyam	15	9860599496	Rautahat	Male
Edit Copy Delete	126	ujjwal	19	9860599495	bharatpur	Male
Edit Copy Delete	128	ujjwal	12	9860599491	pkr	Male

Extra options

	id	name	age	mobile	address	gender
Edit Copy Delete	5	ram	10	9800000000	KTM	Male

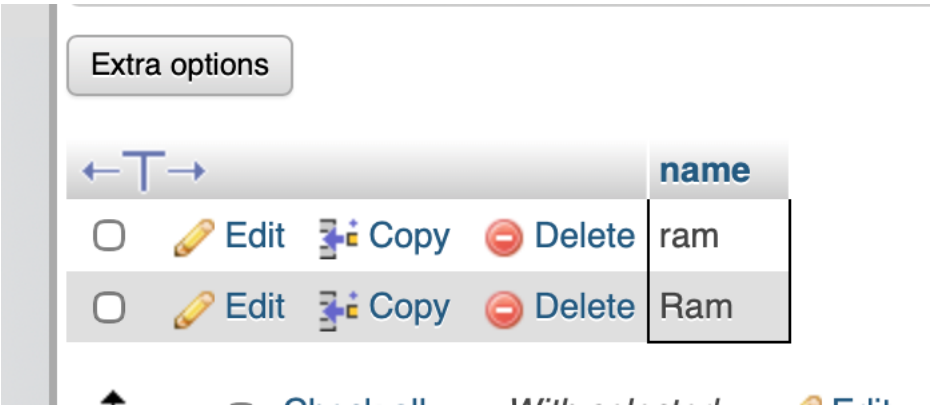
Check all *With selected:* Edit Copy Delete Export

Q8. Select info where id > 5

SELECT * FROM student WHERE id>5;

Q9. Select info where name is 'Ram' and address is 'Ktm'

SELECT * FROM `student` WHERE name='ram' AND address='ktm';

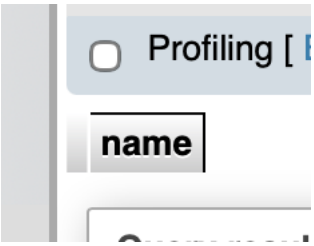


Q10. Display name starting with 'R' only

SELECT name FROM student WHERE name LIKE ‘r%';

Q12. Display info of Female students

```
SELECT * FROM student WHERE gender='famale';
```



Q13. Display name which last letter is ‘e'

```
SELECT name FROM `student` WHERE name LIKE ‘%e';
```

			id	name	age	mobile	address	gender
Edit	Copy	Delete	5	ram	10	9800000000	KTM	Male
Edit	Copy	Delete	123	ujjwal	18	9860599493	chitwan	Male
Edit	Copy	Delete	124	Ram	13	9860599494	gaur	Male
Edit	Copy	Delete	125	shyam	15	9860599496	Rautahat	Male
Edit	Copy	Delete	126	ujjwal	19	9860599495	bharatpur	Male
Edit	Copy	Delete	128	ujjwal	12	9860599491	pkr	Male

			name	age
Edit	Copy	Delete	ujjwal	19

Check all With selected: Edit

Q11. Display info of Male students

```
SELECT * FROM student WHERE gender='male';
```

Q14. Display name and age whose age is 19

			id	name	age	mobile	address	gender
Edit	Copy	Delete	5	ram	10	9800000000	KTM	Male

```
SELECT name, age FROM `student` WHERE age=19;
```

			age
Edit	Copy	Delete	10
Edit	Copy	Delete	18
Edit	Copy	Delete	13
Edit	Copy	Delete	15
Edit	Copy	Delete	19
Edit	Copy	Delete	12

Q15. Display info of student whose mobile has '0' at the end

```
SELECT * FROM `student` WHERE mobile%10=0;
```

Q16. Display distinct age

```
SELECT DISTINCT age FROM `student`;
```

			id	name	age	mobile	address	gender
Edit	Copy	Delete	123	ujjwal	18	9860599493	chitwan	Male
Edit	Copy	Delete	126	ujjwal	19	9860599495	bharatpur	Male

			id	name	age	mobile	address	gender
Edit	Copy	Delete	125	shyam	15	9860599496	Rautahat	Male

			id	name	age	mobile	address	gender
Edit	Copy	Delete	123	ujjwal	18	9860599493	chitwan	Male

Q17. Display age between 15 and 20

```
SELECT * FROM `student`WHERE age>15 AND age<20;
```

			id	name	age	mobile	address	gender
Edit	Copy	Delete	5	ram	10	9800000000	KTM	Male
Edit	Copy	Delete	124	Ram	13	9860599494	gaur	Male
Edit	Copy	Delete	125	shyam	15	9860599496	Rautahat	Male
Edit	Copy	Delete	126	ujjwal	19	9860599495	bharatpur	Male
Edit	Copy	Delete	128	ujjwal	12	9860599491	pkr	Male

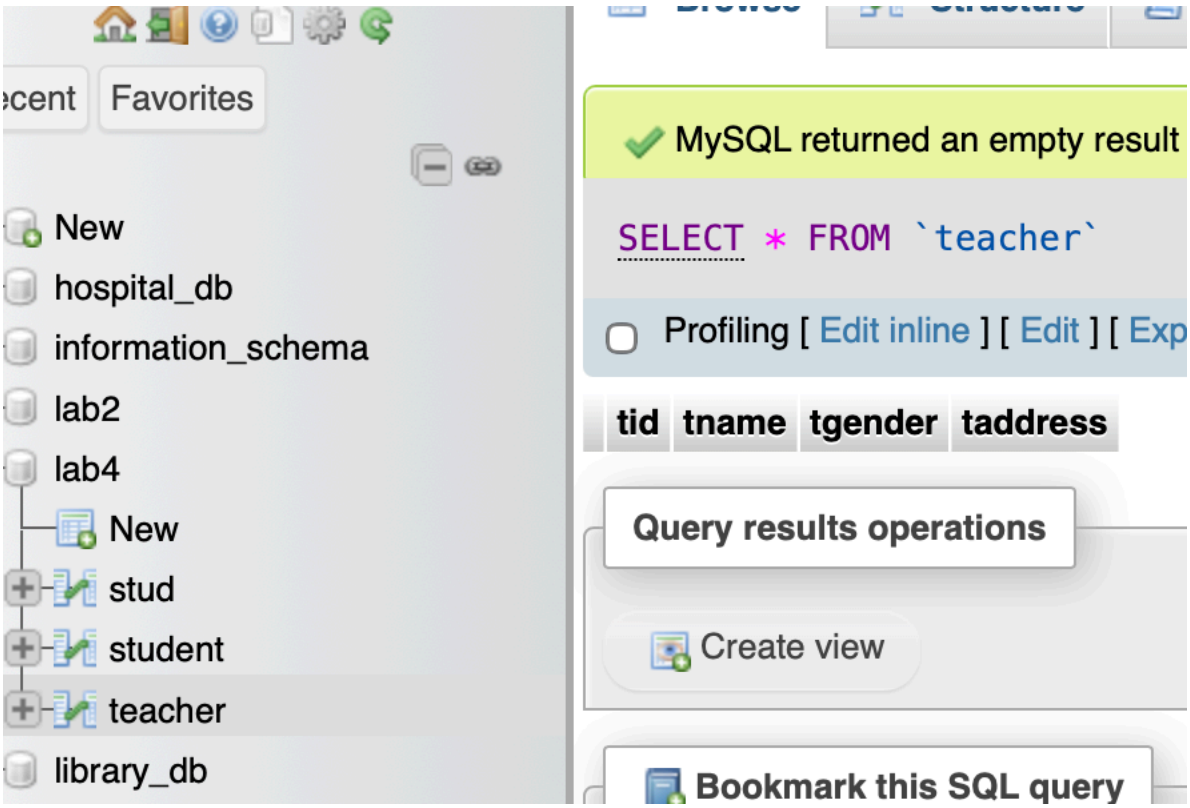
Q18. Display student info where address starts with 'r' and ends with ‘t'

```
SELECT * FROM `student` WHERE address LIKE ‘r%t';
```

			id	name	age	mobile	address	gender	marks
Edit	Copy	Delete	5	ram	10	9800000000	KTM	Male	*NULL*
Edit	Copy	Delete	123	ujjwal	18	9860599493	chitwan	Male	*NULL*
Edit	Copy	Delete	124	Ram	13	9860599494	gaur	Male	*NULL*
Edit	Copy	Delete	125	shyam	15	9860599496	Rautahat	Male	*NULL*
Edit	Copy	Delete	126	ujjwal	19	9860599495	bharatpur	Male	*NULL*
Edit	Copy	Delete	128	ujjwal	12	9860599491	pkr	Male	*NULL*

Q19. Display student info where address ends with 'u', 'n', or 'a'

```
SELECT *
FROM student
```




```
WHERE address REGEXP '[una]$';
```

Q20. Display student info where name is NOT starting with 'A', 'C', or 'D'

```
SELECT *
FROM student
WHERE address REGEXP '^[^acd]';
```

Q21. Add column marks with CHECK constraint marks > 50

```
ALTER TABLE student
ADD COLUMN marks INT CHECK (marks > 50);
```

Q22. Create table teacher

```
CREATE TABLE teacher(
    tid int,
```

```
    tname varchar(30),
    tgender varchar(30),
    taddress varchar(30)
    )
```

Table	Create Table
teacher	CREATE TABLE `teacher` (`tid` int(11) NOT NULL, `tname` varchar(30) DEFAULT NULL, `tgender` varchar(30) DEFAULT NULL, `taddress` varchar(30) DEFAULT 'ktm', PRIMARY KEY (`tid`), UNIQUE KEY `uq_tname` (`tname`), CONSTRAINT `chk_tgender` CHECK (`tgender` in ('male','female'))) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	**tid**	int(11)			No	*None*			Change	Drop	More
☐	2	**tname**	varchar(30)	utf8mb4_general_ci		Yes	*NULL*			Change	Drop	More
☐	3	**tgender**	varchar(30)	utf8mb4_general_ci		Yes	*NULL*			Change	Drop	More
☐	4	**taddress**	varchar(30)	utf8mb4_general_ci		Yes	ktm			Change	Drop	More

Q23. Add constraints to teacher table using ALTER TABLE

```
ALTER TABLE teacher
```

```
ADD CONSTRAINT pk_tid PRIMARY KEY (tid);

ALTER TABLE teacher
ADD CONSTRAINT uq_tname UNIQUE (tname);
```

Table	Create Table
stud	CREATE TABLE `stud` (`sid` int(11) NOT NULL, `id` int(11) DEFAULT NULL, `fee_pay` decimal(10,0) DEFAULT NULL, `fee_due` decimal(10,0) DEFAULT NULL, PRIMARY KEY (`sid`), KEY `id` (`id`)) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci

```
ALTER TABLE teacher
ADD CONSTRAINT chk_tgender
CHECK (tgender IN ('male', 'female'));

ALTER TABLE teacher
ALTER taddress SET DEFAULT 'ktm';
```

Q24. Drop PRIMARY KEY

```
ALTER TABLE teacher DROP PRIMARY KEY;
```

Q25. Drop FOREIGN KEY

```
ALTER TABLE stud DROP FOREIGN KEY stud_ibfk_1;
```